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Exercise 2 Questions

1. Why is it important to validate email format on the backend?

Backend email validation ensures that only correctly formatted email addresses are stored in the database. It prevents invalid supplier information, reduces data entry errors, and ensures the system can communicate with suppliers using valid contact details.

2. What would happen if a product is created with a supplier_id that doesn't exist?

The database will reject the product creation because the supplier_id is a foreign key. This maintains data integrity by preventing products from being linked to suppliers that do not exist.

Exercise 3 Questions

1. How would you handle a scenario where some products in the category would fall below the minimum price?

The system should reject the bulk update operation or skip those products and return an error message explaining that the discount would reduce the price below the allowed minimum of KSh 100.

2. Should this endpoint be available to all users or only administrators?

The endpoint should only be available to administrators because changing prices affects business operations and could cause financial losses if unauthorized users modify prices.

Exercise 4 Questions

1. Why would it be problematic to allow negative stock adjustments without validation?

Allowing negative stock adjustments could result in incorrect inventory records, negative stock values, and inaccurate information about available products.

2. How would you handle concurrent stock adjustments?

The system should use database transactions and locking mechanisms to prevent two users from updating the same product stock at the same time and causing incorrect stock values.

Exercise 5 Question

Why is it important to have a consistent error response format across all endpoints?

A consistent error format makes it easier for developers and users to understand errors. It improves debugging, allows frontend applications to handle errors properly, and provides a better user experience.